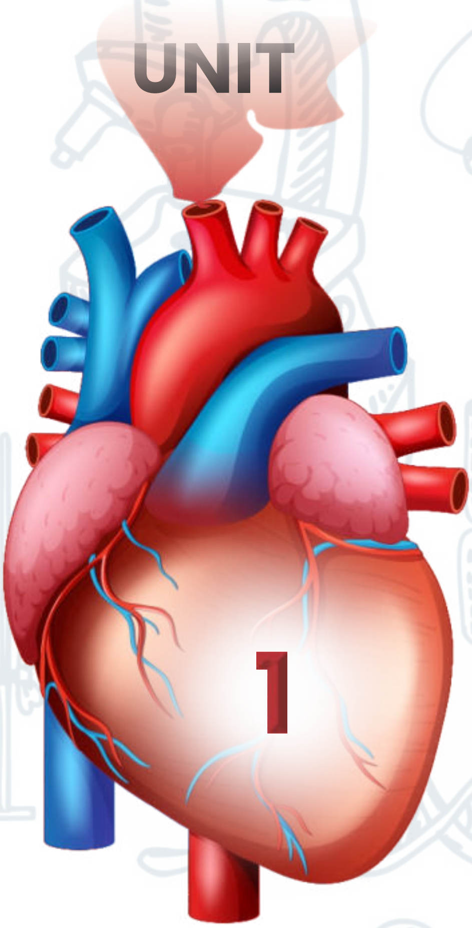
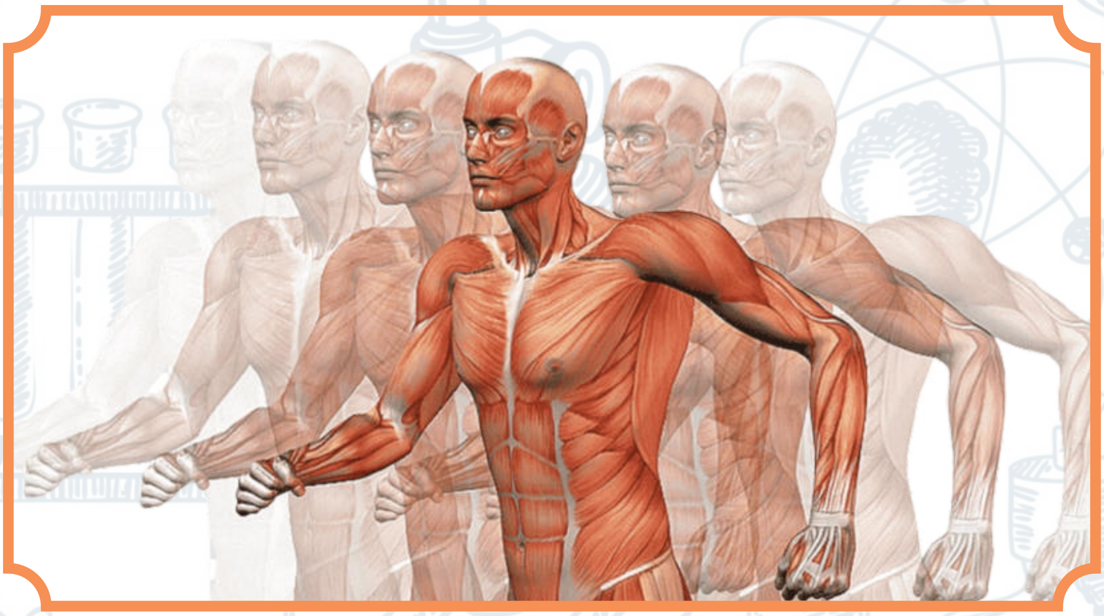


BIOLOGICAL MOLECULES



BIOCHEMISTRY

Definition: "it is the branch of biology which deaths with the structure, composition, functionand chemical processes of molecules and compounds that are present in the bodies of a livingorganism."

WATER:

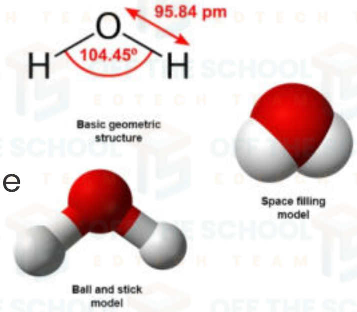
Water is essential for life. There is no existence of life without water. Allah the Almighty has created all living organisms from the water. The bodies of living organisms contain 70 % to 90% water. It is essential for the existence of protoplasm because protoplasm cannot survive if itswater content is reduced as low as 10 percent. Because of its immense importance water ispresent in sufficient amounts in most of the places on earth.

Chemical and Physical Properties Of Water:

The chemical and physical properties of water are so designed that they are absolutely importantfor the vital processes of life.

i. Polar nature of water:

Water is a polar molecule. The oxygen end of the molecule is electronegative bearing a partialnegative charge and the hydrogen end is electropositive bearing a partial positive charge. Watermolecules are bonded to one another through hydrogen bonds. Hydrogen bonds are much weakerthan covalent bonds but they still cause water molecules to remain attached together.



ii. Universal solvent:

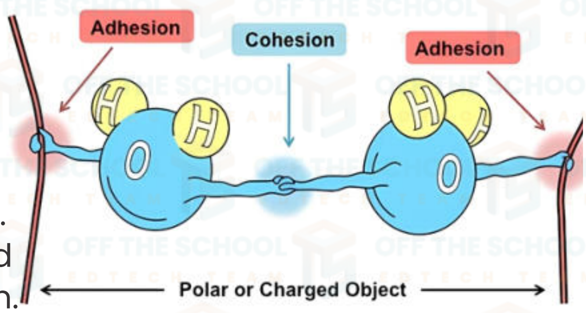
Due to the polar nature of water, it dissolves almost all types of polar substances and is thereforeregarded as a universal solvent. This facilitates chemical reactions both inside and outside the living cell. Water supplies the medium for most chemical responses in cells.

Example:

When sodium chloride is put into water, the electronegative ends of water molecules are attracted to the sodium ions and the electropositive ends of water molecules are attracted to thechloride ions. As a result, the sodium and chloride ions separate and dissolve in water. Waterdissolves all minerals present in soil which are absorbed by plant roots and transported to other tissues.

iii. Cohesive and adhesive force of water:

The water molecules remain attached together and do not separate because of hydrogenbonding. This develops cohesive force among them and therefore water flows freely withoutbreaking apart. Water molecules also adhere (stick) to surfaces. It can fill a tubular vessel and still flows so that dissolved molecules are evenly contributed throughout a system.



iv. High specific heat:

Water has high specific heat. Specific heat is the amount of heat energy required to raise thetemperature of one gram of water by one degree Celsius. It means that water absorbs or releaseslarge quantities of heat energy with little change in temperature. This is why the temperature of the water rises and falls more slowly as compared to other liquids. This property helps theorganisms to maintain body mental temperature and protect themfrom rapid temperature changes.

v. High heat of vaporization:

Water also has a high heat of vapourization. The heat of vapourization helps animals and plants to get rid of excess body heat during sweating and transpiration respectively. The presence of hydrogen bonds among the water molecules cause water to remain liquid rather than change to ice or steam. Without hydrogen bonds, water would boil at -80 o C and would freeze at -100 o C. In such conditions, life for living organisms would become impossible.

vi. Water expands at low temperature:

Water has a unique property, as it expands when the temperature falls below 4°C. Water is mostheavy at 4 o C. Therefore ice (solid water) is less dense than liquid water and this is the reason that ice floats in liquid water. The water body freezes on the surface at low temperature.

vii. High surface tension:

Water has a high surface tension. In living cells this part of surface tension allows a thin movie of water to cover membranes and to keep them moist.

viii. Ionization of Water:

Water molecules may ionize into hydrogen ions (H^+) and hydroxyl ions (OH^-). Very few molecules out of a very large number may ionize. The presence of ions is important for the normal functioning of enzymes.

CONDENSATION	HYDROLYSIS
The process by which two monomers are joined is called; condensation;	A process during which polymers are broken dawn into their subunits (monomers) by the addition of 1-120 called Hydrolysis.
In this process two monomers join together when a hydroxyl(OH)group is removed from onemonomer and a hydrogen (-H) isremoved from other monomer.	In this process a polyomer is broken down by the addition of a hydroxyl(OH) and a hydrogen from water molecule.

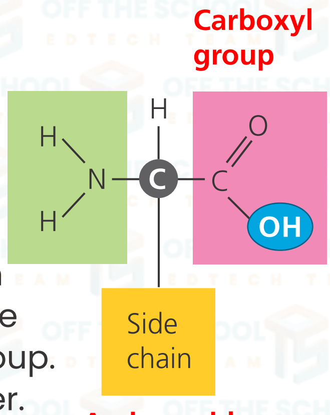
PROTEIN

Proteins are macromolecules (polymers) formed of units (monomers) called amino acids. A large number of amino acids are known. Of these, only twenty different types of amino acidscombine in different numbers and different sequences forming hundreds and thousands ofdifferent types of protein molecules.

Proteins are the most abundant organic compounds of the cell. They play the most importantrole in cellular functions. They contain elements of carbon, hydrogen, oxygen, and nitrogen.Some proteins also contain sulfur.

STRUCTURE OF AMINO ACIDS

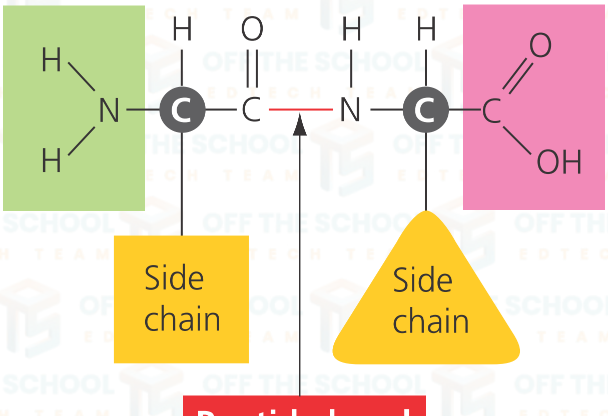
Amino acids are carboxylic acids having amino groups. Each amino acid has a central carbon atom called alpha carbon. There are four different groups attached to the alpha carbon. These are an amino group, carboxyl group, hydrogen of alpha carbon and an R group. The former three groups attached to the alpha carbon are constant members and are present in all amino acids while the fourth one i.e. R group is variable. It is either a hydrogen or alkyl group. Due to this variable, the amino acids are different from one another.



Amino acid

PEPTIDE BONDS

A bond called peptide bond links amino acids in a protein molecule to each other. The peptide bond is formed between an amino group of one amino acid and a carboxyl group of anotheramino acid. This is a dehydration or condensation reaction in which one water molecule isformed.



Dipeptide

A chain containing three amino acids and two peptide bonds is known as dipeptide chain.

Tripeptide

A chain with four amino acids and three peptide bonds is called tripeptide chain|

Polypeptide

A chain with many amino acids and many peptide bonds is called polypeptide. Most protein molecules are usually formed of two or many polypeptide chains e.g. haemoglobinand insulin.

Example

Haemoglobin is an oxygen-carrying protein in the red blood cells which consists of four polypeptide chains while an insulin molecule is consists of two polypeptide chains. The number and sequence of amino acids in a protein molecule is highly specific for its normal function. If an amino acid is not occupying its specific position in a protein molecule it will fail to perform its function.

THE SHAPE OF PROTEIN

As regards the shape, proteins are classified into two types; fibrous proteins and globular proteins

a. Fibrous proteins

The molecules of fibrous proteins are composed of one or more polypeptide chains, which are linearly arranged in the form of fibers. They are water-insoluble. Some of these may form sheet-like structures.

Examples of fibrous proteins

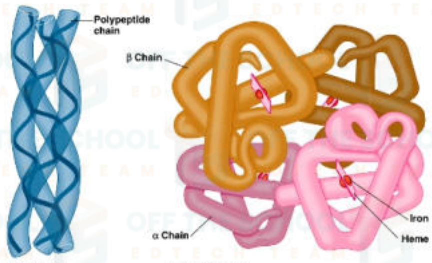
Keratin found in hairs, nails, fur, outer skin, myosin present in muscle cells, collagen which isthe most abundant protein in higher vertebrates found in skin, ligaments, tendons, bones and inthe cornea of the eyes.

b. Globular Proteins

Globular proteins, as the name indicates, are globular or spherical in shape due to the folding ofpolypeptide chains. They are usually water-soluble.

Examples of globular proteins

Haemoglobin, albumen of egg white, enzymes, antibodies and the proteins of cell membranes.



(a) Collagen (b) Hemoglobin

LEVELS OF STRUCTURE (ORGANIZATION)

There are four levels of organization of protein molecules. This is because each type of

polypeptide chain bends, folds and twists in a particular way within a protein molecule. This gives protein molecule a characteristic structure that classifies protein into four different types.

A. PRIMARY STRUCTURE

A polypeptide chain having a linear sequence of amino acids is called the primary structure

.B. SECONDARY STRUCTURE

When a polypeptide chain of amino acids become spirally coiled, the structure is called a secondary structure of the protein.

C. TERTIARY STRUCTURE

When the secondary structure of a protein is arranged into a three-dimensional structure, it iscalled a tertiary structure.

D. QUATERNARY STRUCTURE

When two or more polypeptide chains are arranged into a large-sized molecule, it is called a quaternary structure.

Functions of Proteins

- Proteins perform the most important functions in the life of living organisms.

Proteins are the structural and building materials of cellular membranes called lipoprotein membranes.

- All enzymes are proteins. They speed up biochemical reactions inside the body of living or ganisms.
- The digestive enzymes are important for the process of digestion. Without their presence, food cannot be digested.
- Some hormones such as insulin are proteins which regulate biochemical processes.
- Myosin and actin fibers play an important role in the contraction of muscles and movements.
- Haemoglobin is an oxygen-carrying protein of red blood cells.
- In animals' proteins form most structures such as skin, nails, hairs, claws, hooves etc.
- In plants, proteins are stored in most seeds for the future need of the embryos e.g. bean, pulses, pea etc.

CARBOHYDRATES

It is a group of organic compounds having carbon, oxygen and hydrogen, in which hydrogen and oxygen are mostly found in the same ratio as in water i.e. 2.1 and thus called Hydrated carbon They are found about 1 % by weight and generally called Sugars or saccharides due to their sweet taste except polysaccharides.

CLASSIFICA TION OF CARBOHYDRA TES

The carbohydrates can be classified into following groups on the basis of number of monomers.

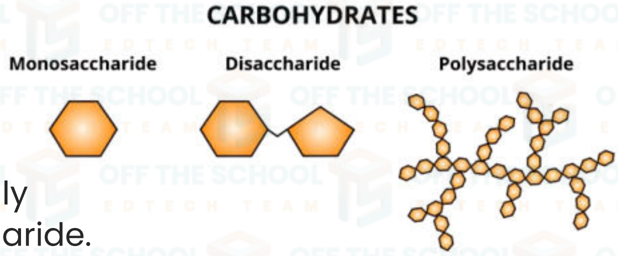
1 . Monosaccharide
2. Oligosaccharides
3. Polysaccharides.

(1) MONOSACCHARIDES

These are called "Simple Sugars", because they cannot be hydrolysed further into simple sugars.Their general formula is "C n H2 n O nThey are present in various fruits and vegetables. E.g: Glucose, Galactose, Fructose and Riboseetc. Monosaccharide can be sub-classified according to umber of carbon atom present in eachmolecule. They may be triose, (Glycerose), tetrose (erythrose), pentose, (ribose), hexone(glucose) or heptose (Glucoheptose) having 3,4,5 ,6 and 7 carbon atoms respectively.

(2) OLIGOSACCHARIDES

These carbohydrates yield 2to 1 0 monosaccharides nolecules on hydrolysisDisaccharides are themost common and abundant carbohydrates of oligosaccharides.These sugars are comparatively less sweet in taste, and less soluble in water.E.g: Maltose, Sucrose and lactose etc.



(3) POLYSACCHARIDES

These are the most complex and most abundant carbohydrates in nature.They are of high molecular weight carbohydrate which on hydrolysis yield mainly monosaccarides or products related to monosaccharide.

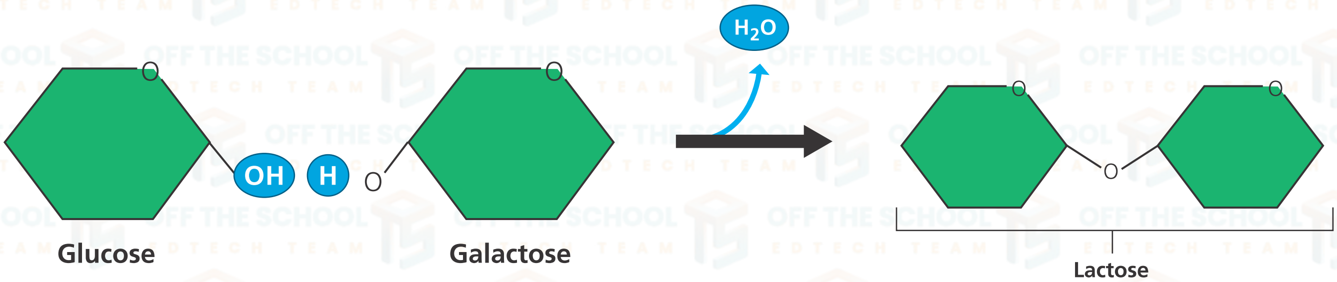
These sugars are formed by the condensation of hundreds of thousands of monosaccharide units.They are tasteless and only sparingly souble in 1-120. E.g: Strach, cellulose Glycogen , Dextrin Agar, pectin and Chitin etc

Disaccharides:

The oligosaccharides that are hydrolyzed into two simple units are called disaccharides, those hydrolyzed into three units are trisaccharides and so on. Disaccharides are the most common oligosaccharides.

Example:

Common disaccharides are sucrose, lactose, and maltose. Sucrose is present in sugarcane and is hydrolyzed into glucose and fructose.



The covalent bond that is formed between two monosaccharide units is called a glycosidic bond. Lactose is found in milk that contains galactose and glucose.

Lactose + Water $\xrightarrow{\text{(Hydrolase)}}$ Glucose + Galactose

Maltose is disaccharide found in fruits. It is composed of two glucose units and is found in ourdigestive tract as a result of starch digestion.

Maltose + Water $\xrightarrow{\text{(Hydrolase)}}$ Glucose + Glucose

Physical properties of Disaccharide:

Depending on the monosaccharide constituents, disaccharides show a variation of the following properties

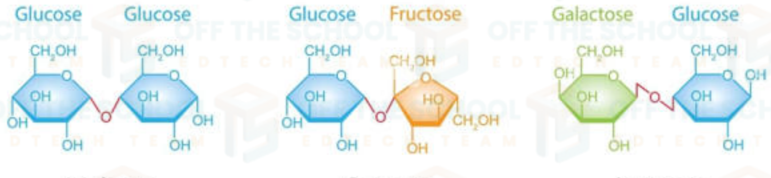
- Crystalline
- Sometimes water-soluble
- Sometimes sweet-tasting and sticky-feeling.

Common Disaccharides:
Some of the common disaccharides are as follows.

1- Sucrose (table sugar, cane sugar, beet sugar):
Sucrose is a white, odorless crystalline powder with a sweet taste. It is best known for its role in human nutrition. Sucrose is made up of glucose (monosaccharide) and fructose (monosaccharide).

2. Lactose (milk sugar):
Lactose is found in milk. Lactose is formed from galactose and glucose. Lactose has a complex molecular structure, and so some people are unable to digest it properly.

3. Maltose:
Maltose (malt sugar) is formed from bonding between two units of glucose. It is used in the making of soft candies such as chocolates and fruit based-treats.



FUNCTIONS OF CARBOHYDRATES
• Carbohydrates are the potential source of energy.
• They act as storage food molecules and also work as an excellent building, protective and supporting structure.
• They also form complex conjugated molecules.
• They are needed to synthesize lubricants and are also needed to prepare the nectar in some flowers.

LIPIDS
These are naturally occurring compounds, which are insoluble in water but soluble in organic solvents. They contain carbon, hydrogen and oxygen like carbohydrates rate but in much lesser ratio of oxygen than carbohydrates. These biomolecules are widely distributed among plants and animals.

CLASSIFICATION OF LIPIDS
Following are the important groups of lipids.
1. Acylglycerol (fats and oil)
2. Waxes
3. Phospholipids'.
4. Terpenoids.

(1) ACYLGLYCEROL (FATS AND OIL
These are found in animals and plants, provide energy for different metabolic activates and are very rich in chemical energy.They are composed of glycerol and fatty acids. The most widely spread acylglycerol triacyl glycerol, also called triglycerides or natural lipids.There are two types of acylgycerol.

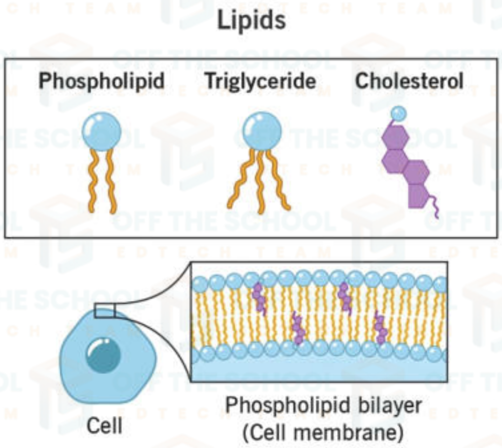
(A) Saturated Acylglycerol
They contain no double bond. They melt at higher temperature than unsaturated acylglycerols. Lipids containing saturated lipids. E.g Butter, Ghee etc

(B) Unsated Acylgl Glycerol
They contain unsaturated fatty acids i.e they contain one or more than one double bond between carbon atom(C=C-).They are liquid at ordinary temperature. They are found in plant also called Oil E.g: linolin found in cotton seeds etc.

(2) WAXES
Chemically waxes are mixtures of long chain alkanes and alcohols. Ketones and esters of long chain fathy acidsWaxes are widespread as protective coatings of fruits and leaves some insects also secrete wax. Waxes protect plants form water loss and abrasive damage.They also providewater barrier for insects, birds and animals etc.

(3) PHOSPHOLIPIDS
It is most important class of lipids from biological point of view and is similar to riacylglycerol or an oil except that one fatty acid is replaced by phosphate group.The molecule of phospholipids consist of two ends, which are called hydrophilic (water loving end (head) and hydrophobic (water fearing)end (Tail).These are frequently associated with membranes and are related to vital functions such as regulation of cell permeability and transport process.

(1) TERPENOIDS
It is large and important class of lipids containing Isoprenoid unit (C5H8).They help in oxidation reduction process, act as components of essential oils of plants and also found in cell membrances cholesterol



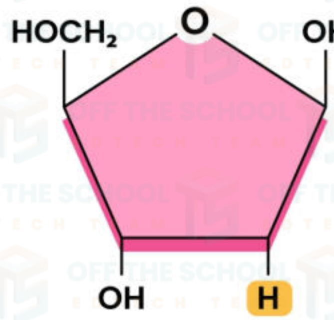
.SUB-CLASSES OF LIPIDS
1. Terpenes
2. Steroids
3. Carotenoids.
4. Prostaglandins

NUCLEIC ACIDS
Acids Were First Isolated In 1870 By an Austrian Physician Fridrich Micscher fromthe nuclei of pus cells. These bio molecules are acidic in nature and present in the nucleus.

TYPES OF NUCLEIC ACIDS
Nucleic acids are of two types.
1 . Deoxyribonucleic acid or DNA
2. Ribonucleic acid or RNA

CHEMICAL NATURE OF NUCLEIC ACID

Nucleic acids are complex substances. They are polymers of units called nucleotides. DNA is made up of deoxyribonucleotides, while RNA is composed of ribo nucleotides.

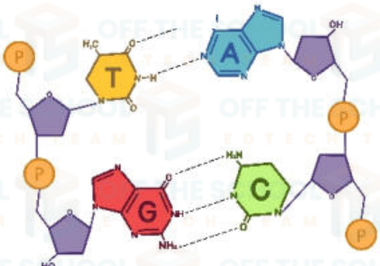


STRUCTURE OF NUCLEOTIDE

Each nucleotide is made of three subunits
a) 5-carbon monosaccharide (a pentose sugar)
b) Nitrogen containing base.
c) Phosphoric acid.

(A) PENTOSE SUGAR

Pentose sugar in RNA is ribose, while in DNA it is deoxyribose.

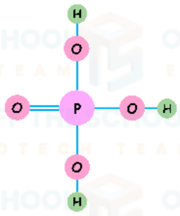


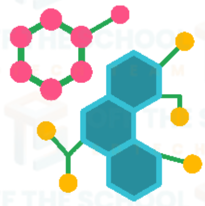
(B) NITROGENOUS BASE

Nitrogenous bases are of two types
(I) PYRIMIDINES (SINGLE RINGED): These are cytosine (abbreviated as C), thymine(abbreviated as T), and uracil (abbreviated as U).
(II) PURINES (DOUBLE RINGED): These are adenine (abbreviated as A) and guanine(abbreviated as G).

C) PHOSPHORIC ACID

Phosphoric acid (H3P04) has the ability to develop ester linkage with OH group of pentose sugar.





FORMATION OF NUCLEOTIDE

Formation of nucleotide takes place in two steps. First the mitrogenous base combines with pentose sugar at its first carbon to form a "Nucleoside". In second step the phosphoric acid combines with the 5th carbon of pentose sugar to form a "Nucleotide".

(A) MONONUCLEOTIDES

They exist singly in the cell or as a part of other moleculeThese are not the part of DNA or RNA and some of these have extra phosphate groups e.g ATP.

B) DINCULEOTIDES

These nucleotides are covalently bounded together and usually act as co-enzymes E.g NAD (Nicotinamide dinucleotide).

(C) POL YNUCLEOTIDES

Nucleotides are found in the nucleic acid as "'Polynucleotide" and they have a variety of role in living organisms.

DEOXYRIBONUCLEIC ACID (DNA):

In 1962, James Watson, Francis Crick, and Maurice Wilkins jointly received the Nobel Prize in physiology or medicine for their 1953 determination of the structure of deoxyribonucleic acid (DNA).

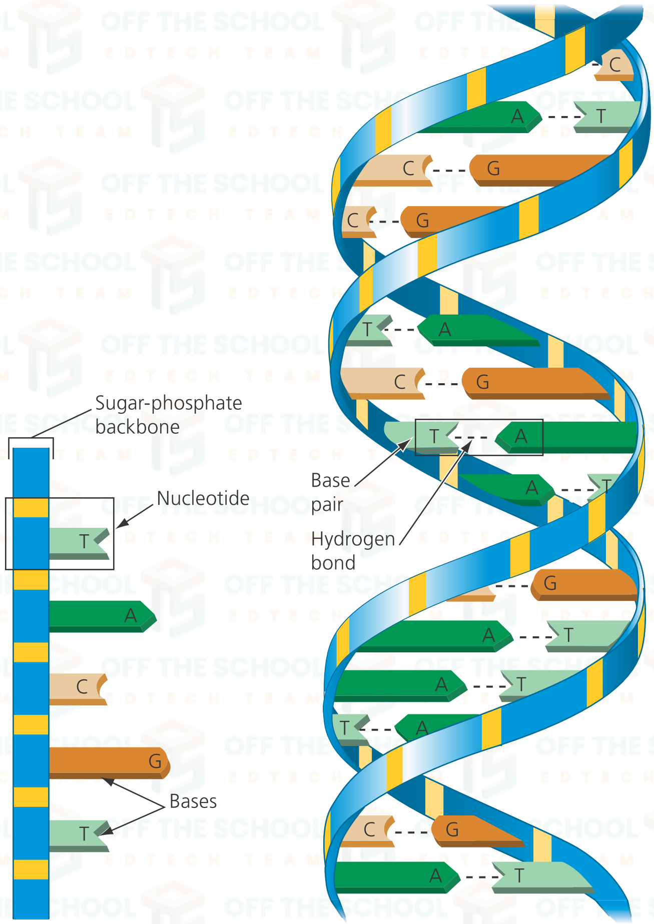
According to this model, the DNA molecule is a double helix.A DNA molecule consists of two strands. The two strands twist about one another in the form of a double helix.

DNA contains pentose sugar as deoxyribose. It is formed of four different types of nucleotides. These nucleotides are named after the base present in them they are Adenine deoxyribonucleotide, Guanine deoxyribonucleotide. Thymine deoxyribonucleotide and Cytosine deoxyribonucleotide.

These four types of nucleotides are used as building blocks of DNA molecule. The nucleotides in the DNA molecule are bonded to one another in such a manner that the sugar of one nucleotide is linked to the phosphate group of the next one.Two strands run in opposite direction to each other in the double helix. They are held together by hydrogen bonds between purine and pyrimidine bases. Thymine in one strand is always paired with adenine in the opposite strand and guanine is always paired with cytosine.

There are two hydrogen bonds between adenine and thymine and three hydrogen bonds between guanine and cytosine.

▼ **The structure of DNA.** The base pairing in a DNA molecule is specific: A always pairs with T; G always pairs with C.



(a) DNA strand (polynucleotide)

(b) Double helix (two polynucleotide strands)

CONJUGATED MOLECULES:

"A conjugated molecule is formed by the combination of two different molecules belonging to different categories."

Example:

When a carbohydrate molecule combines with protein, a conjugated molecule called glycoprotein is formed. Other examples are nucleoproteins, glycolipids, and lipoproteins.

Types of conjugated molecules:

1. Lipoproteins:

Lipoprotein is formed by the combination of lipids and proteins. Lipoproteins are the basicstructural framework of the plasma membrane and all other types of membranes in the cell.

2. Nucleoproteins:

Nucleoprotein is formed by the combination of nucleic acids with proteins. A eukaryotic chromosome is basically a nucleoprotein that is formed by the DNA and protein. These are slightly acidic and soluble in water.

3. Glycolipid:

Glycolipid is formed by the combination of carbohydrates and lipids. Glycolipids are an important component of brain and plasma membrane.

4. Glycoprotein:

Glycoproteins are formed by the combination of carbohydrates and proteins. Glycoproteins are an integral component of the plasma membrane. They are also present in egg albumin.